

INSTAGRAM USER ANALYTICS SQL PROJECT



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Introduction

In today's digital landscape, understanding user engagement is crucial for the success of any social media platform. This project focuses on analyzing Instagram user interactions using SQL to extract valuable insights. By leveraging MySQL Workbench, we examine user behavior, posting activity, hashtag trends, and potential bot detection. The insights derived from this analysis assist the product team in enhancing user experience, the marketing team in optimizing campaigns, and investors in evaluating platform engagement. Through data-driven decision-making, this project aims to support Instagram's growth and strategic improvements.

Project Description:

This project aims to analyze Instagram user behavior and engagement using SQL. By extracting and interpreting key data points, we gain insights into user activity, content trends, and platform interactions.

The analysis benefits multiple teams:

- The product team can refine features based on user engagement patterns.
- The marketing team can enhance campaigns by identifying loyal and inactive users.
- Investors can evaluate platform growth and detect potential bot activity.

Using SQL queries in MySQL Workbench, this project provides data-driven insights to support informed decision-making and optimize Instagram's overall strategy.

Approach :

- **Database Creation:**

1. Used DDL (Data Definition Language) queries to create necessary tables in the MySQL database.
2. Used DML (Data Manipulation Language) queries to insert the provided data into the tables.
3. Ensured data integrity and consistency for accurate analysis.

- **Data Extraction & Analysis:**

1. Ran SQL queries in MySQL Workbench to retrieve insights from the database tables.
2. Applied aggregation functions, joins, and filtering techniques to analyze user engagement, posting behavior, and hashtag trends.
3. Identified key metrics such as loyal users, inactive users, contest winners, and bot activity based on query results.

- **Insight Generation & Reporting:**

1. Interpreted query outputs to provide meaningful insights for the product, marketing, and investor teams.
2. Compiled findings into a structured report with SQL queries, outputs, and visualizations for better understanding.

Tech-Stack Used:

- **MySQL Database & MySQL Workbench (Version: X.X)**

1. Chosen for its reliability, scalability, and ease of use for managing structured data.
2. MySQL Workbench provides a graphical interface for writing and executing SQL queries efficiently.

- **SQL (Structured Query Language)**

1. Used for creating databases (DDL), inserting data (DML), and extracting insights through queries.
 2. Essential for performing data analysis, aggregations, and filtering to derive meaningful insights.
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Insights:

1. Marketing Analysis:

1. Rewarding Most Loyal Users:

Identify the five oldest users on Instagram from the provided database.

```
select * from users
order by created_at asc
limit 5 ;
```

	id	username	created_at
	80	Darby_Herzog	2016-05-06 00:14:21
	67	Emilio_Bernier52	2016-05-06 13:04:30
	63	Elenor88	2016-05-08 01:30:41
	95	Nicole71	2016-05-09 17:30:22
	38	Jordyn.Jacobson2	2016-05-14 07:56:26
	NULL	NULL	NULL

2.Inactive User Engagement:

Identify users who have never posted a single photo on Instagram..

```
select * from users u left join  photos p on
p.user_id = u.id
where p.image_url is null;
```

id	username	created_at	id	image_url	user_id	created_dat
5	Aniya_Hackett	2016-12-07 01:04:39	NULL	NULL	NULL	NULL
7	Kasandra_Homenick	2016-12-12 06:50:08	NULL	NULL	NULL	NULL
14	Jadyn81	2017-02-06 23:29:16	NULL	NULL	NULL	NULL
21	Rocio33	2017-01-23 11:51:15	NULL	NULL	NULL	NULL
24	Maxwell.Halvorson	2017-04-18 02:32:44	NULL	NULL	NULL	NULL
25	Tierra.Trantow	2016-10-03 12:49:21	NULL	NULL	NULL	NULL
34	Pearl7	2016-07-08 21:42:01	NULL	NULL	NULL	NULL
36	Ollie_Ledner37	2016-08-04 15:42:20	NULL	NULL	NULL	NULL
41	Mckenna17	2016-07-17 17:25:45	NULL	NULL	NULL	NULL
45	David.Osinski47	2017-02-05 21:23:37	NULL	NULL	NULL	NULL

3.Contest Winner Declaration:

Determine the winner of the contest and provide their details to the team.

```
select users.username , photos.id, count(likes.user_id) as Total_likes
from photos inner join likes on
photos.id=likes.photo_id
inner join users on
photos.user_id=users.id
group by photos.id
order by Total_likes DESC
limit 1;
```

	username	id	Total_likes
	Zack_Kemmer93	145	48

4.Hashtag Research:

Identify and suggest the top five most commonly used hashtags on the platform.

```
select tags.tag_name,count(*) as total_tags from photo_tags join tags on
photo_tags.tag_id=tags.id
group by tags.id
order by total_tags desc
limit 5;
```

tag_name	total_tags
smile	59
beach	42
party	39
fun	38
concert	24

5.Ad Campaign Launch:

Determine the day of the week when most users register on Instagram. Provide insights on when to schedule an ad campaign.

```
select dayname(created_at) as day, count(*) as total_reg
from users
group by day
order by total_reg desc
limit 1;
```

day	total_reg
Thursday	16

2.Investor Metrics:

1.User Engagement

Calculate the average number of posts per user on Instagram. Also, provide the total number of photos on Instagram divided by the total number of users.

```
SELECT (SELECT Count(id)
FROM   photos) / (SELECT Count(DISTINCT user_id)
FROM   photos) AS Average_posts_per_User,
(SELECT Count(id)
FROM   photos) / (SELECT Count(id)
FROM   users) AS Ratio_of_Total_Posts_to_Total_Users;
```

Average_posts_per_User	Ratio_of_Total_Posts_to_Total_Users
3.4730	2.5700

2.Bots & Fake Accounts:

Identify users (potential bots) who have liked every single photo on the site, as this is not typically possible for a normal user.

```
SELECT id,username
FROM users
WHERE id IN (SELECT user_id
              FROM likes
              GROUP BY user_id
              HAVING Count(user_id) = (SELECT Count(id)
                                       FROM photos));
```

id	username
5	Aniya_Hackett
14	Jadyn81
21	Rodio33
24	Maxwell.Halvorson
36	Ollie_Ledner37
41	Mckenna17
54	Duane60
57	Julien_Schmidt
66	Mike.Auer39
71	Nia_Haag
75	Leslie67
76	Janelle.Nikolaus81
91	Bethany20

Results...

This project provided valuable hands-on experience in SQL and data analysis by extracting meaningful insights from Instagram user data. I successfully created and managed a database using DDL and DML queries in MySQL Workbench, enabling efficient data organization. Through various SQL queries, I analyzed key metrics such as user engagement, inactive users, contest winners, hashtag trends, and bot activity, offering insights that can aid in marketing strategies, product improvements, and investor evaluations. This project enhanced my SQL skills, problem-solving abilities, and data interpretation expertise, reinforcing the significance of data-driven decision-making in real-world business scenarios.



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THANK YOU !!